

Current updates on viral infections affecting tilapia

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ABSTRACT

Tilapia aquaculture is a commercially important and lucrative trade with global production predicted to increase in the future. Owing to the robust and disease-resistant nature of tilapia, tilapia aquaculture has expanded and intensified over the years. However, the intensification of tilapia aquaculture has led to the emergence of diseases, whereby viral diseases represent a prominent limiting factor to the growth of aquaculture. This review aims to discuss all viral infections in tilapia, with emphasis on the aetiology, occurrence, transmission, pathology, diagnostic methods, and control measures to mitigate outbreaks of viral diseases in aquaculture. To date, ten viral infectious diseases have been reported globally in various tilapia species reared in aquaculture or laboratory settings. To our knowledge, this review article represents the first instance of reviewing tilapia parvovirus (TiPV) infection in tilapia, in addition to tilapia larvae encephalitis virus (TLEV), infectious spleen and kidney necrosis virus (ISKNV), iridovirus-like agent, novel megalocytivirus, Bohle iridovirus (BIV), lymphocystis disease virus (LCDV), infectious pancreatic necrosis virus (IPNV), nervous necrosis virus (NNV), and tilapia lake virus (TiLV).

1. Introduction

Tilapia is an important aquaculture species with an estimated worldwide production of 6.1 million tonnes worth approximately USD 12.3 billion in 2020, making farmed tilapia production the second highest following carp (Aich et al., 2022; FAO, 2022; MacKinnon et al., 2023; Maulu et al., 2021). Originally native to Africa and the Middle East, tilapia is now cultivated in over 140 countries (Munguti et al., 2022; Wang & Lu, 2016). Tilapia is a common name encompassing over 100 species from the family Cichlidae (Bacharach et al., 2016; Hounmanou et al., 2018) with species from *Oreochromis* being the most commercially important (Khanjani et al., 2022; Ng & Romano, 2013). *Oreochromis* species cultivated worldwide include Nile tilapia (*Oreochromis niloticus*), blue tilapia (*Oreochromis aureus*), Mozambique tilapia (*Oreochromis mossambicus*), redbreast tilapia (*Tilapia rendalii*) and various hybrid tilapia species (*O. aureus* × *O. niloticus*, *O. aureus* × *O. mossambicus*, *O. mossambicus* × *O. niloticus*) (Arumugam et al., 2023; FAO, 2022; Machimbirike et al., 2019).

Tilapia aquaculture is a rapidly expanding trade which can be credited to several advantageous characteristics of tilapia which include robustness, rapid growth, resistance to diseases, adaptability, and tolerance to a wide range of culture conditions (Lakshmi et al., 2023;

MacKinnon et al., 2023; Ng & Romano, 2013; Yáñez et al., 2020). Besides contributing to global food source and security, the expansion of tilapia aquaculture also provides local employment along with domestic and export earnings (MacKinnon et al., 2023; Miao & Wang, 2020; Nicholson et al., 2018). Despite the general belief that tilapia is disease-resistant, the intensification of culture systems to meet the increased demands for tilapia exerts greater levels of stress on the fish (Lakshmi et al., 2023; Maulu et al., 2021; Subasinghe et al., 2019). This in turn increases the risk of infectious disease outbreaks, making infectious disease outbreaks the main factor that impedes the growth of the global tilapia aquaculture industry (Bondad-Reantaso et al., 2005; Lakshmi et al., 2023).

Tilapia are susceptible to various bacterial, viral, fungal, and parasitic infections (AlYahya et al., 2018; Ibrahim, 2020; Pantoja et al., 2012; Ramírez-Paredes et al., 2021; Soto et al., 2012), causing significant production and economic losses (Dong et al., 2023; Fathi et al., 2017). Among the various infectious agents, viruses are undoubtedly the major pathogen that threatens the aquaculture industry due to limited antiviral therapeutics and vaccines (Kibenge et al., 2012; Valero & Cuesta, 2023). For example, Egyptian tilapia farms affected by tilapia lake virus (TiLV) experienced an estimated production loss of 98,000 tonnes, which correlated to an economic loss of USD 100 million (Fathi

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et al., 2017). Another instance of production loss was seen during an infectious spleen and kidney necrosis virus (ISKNV) outbreak in Ghana, where more than 10 tonnes of Nile tilapia were lost per day (Ramírez-Paredes et al., 2021). Hence, there is a need for a comprehensive understanding on the viral infections that affect the tilapia industry so that effective measures for the prevention and control of viral diseases can be implemented (Abdullah et al., 2017).

Presently, there is a lack of up-to-date literature that comprehensively covers all viral infections known to affect tilapia. While Machimbirike et al. (2019) described most viruses known to infect tilapia, TiLV, novel megalocytivirus, and the recently emerging tilapia parvovirus (TiPV) were not discussed. This review aims to describe and discuss all the current viral infections reported in tilapia, including TiPV, TiLV, tilapia larvae encephalitis virus (TLEV), ISKNV, Bohle iridovirus (BIV), lymphocystis disease virus (LCDV), infectious pancreatic necrosis virus (IPNV), nervous necrosis virus (NNV), and iridovirus-like agent.

2. Viral infections affecting tilapia

To date, ten viral infections consisting of seven DNA and three RNA viruses have been reported to infect tilapia, where the timeline of the first emergence of these viral infections and corresponding regions are shown in Fig. 1. As the years advanced from the 1970s to the 2010s, an increasing number of viral diseases have emerged in tilapia, likely attributed to the intensification of aquaculture practices (Debnath et al., 2023). The DNA viruses known to infect tilapia include TiPV (Liu et al., 2020), TLEV (Shlapobersky et al., 2010), ISKNV (Dong et al., 2015), iridovirus-like agent (McGrogan et al., 1998), novel megalocytivirus (Shahin et al., 2021), BIV (Ariel & Owens, 1997), and LCDV (Paperna, 1973). Meanwhile, the RNA viruses known to infect tilapia include NNV (Bigarré et al., 2009), IPNV (Hedrick et al., 1983), and TiLV (Eyngor et al., 2014). The summary of the viral agents affecting tilapia is shown in Table 1, whereas Table 2 depicts the characteristics of viral diseases and diagnostic methods in tilapia. Additionally, Fig. 2 illustrates the characteristics of diseases manifested in tilapia after infection with each DNA virus, while Fig. 3 depicts the disease characteristics observed in tilapia after infection with each RNA virus.

2.1. Tilapia parvovirus

Tilapia parvovirus (TiPV), classified in the family Parvoviridae under the subfamily Hamaparvovirinae and genus Chaphamaparvovirus (ICTV, 2022; Reuter et al., 2022), is a spherical non-enveloped single-stranded DNA (ssDNA) virus with a diameter of 28–32 nm (Liu et al., 2020; Piewbang et al., 2022). Its genome spans 4 268 base pairs, containing two major open reading frames (ORFs) encoding a non-structural protein 1 (NS1) and capsid protein 1 (VP1) (Dong et al., 2022; Liu et al., 2020). First isolated via next-generation sequencing (NGS) of intestinal samples from tilapia and faecal samples from tilapia-fed crocodiles in Hainan, China (Du et al., 2019), TiPV emerged as a significant pathogen

causing fatalities, particularly in cage-cultured tilapia (breed not specified) in Hubei, China, where 60%–70% mortality rates were recorded (Liu et al., 2020).

Recent reports from Thailand indicate TiPV co-infections with tilapia lake virus (TiLV), resulting in mortality rates of 50%–75% (Piewbang et al., 2022; Yamkasem, Tattiyapong, et al., 2021). Co-infections with *Streptococcus agalactiae* and other microbial pathogens have also been documented (Dong et al., 2022; Liu et al., 2020). Experimental intraperitoneal (IP) infection with TiPV resulted in a 90% mortality rate in tilapia (Liu et al., 2020), suggesting its high pathogenicity. TiPV's ability to potentially suppress the immune system, increasing susceptibility to other pathogens, aligns with findings on parvovirus infections in animals (Kotob et al., 2017; Sykes, 2014; Thaiwong et al., 2016).

Clinical signs of TiPV-infected fish include lethargy, abnormal swimming behaviour, anorexia, exophthalmia, ocular lesions, and body haemorrhages (Liu et al., 2020). TiPV manifests systemically in various organs, including the pancreas, musculature, intestine, eyes, gills, heart, brain, spleen, kidney, liver, and mucus, detected through quantitative polymerase chain reaction (qPCR) or *in situ* hybridisation (ISH) (Liu et al., 2020; Piewbang et al., 2022; Yamkasem, Tattiyapong, et al., 2021). Notably, TiPV often targets the kidney and spleen most prominently (Liu et al., 2020; Yamkasem, Tattiyapong, et al., 2021). Interestingly, Piewbang et al. (2022) and Yamkasem, Tattiyapong, et al. (2021) have both reported that TiLV was found co-localised with TiPV in the kidney and spleen, indicating both viruses target similar organs. As such, care should be taken while studying the histopathology of TiPV infection, as the local histopathological lesions observed may not be exclusively caused by TiPV due to the TiLV co-localisation.

Recently, Cowdry type A inclusion bodies (CAIB) in the pancreas of moribund Nile tilapia were proposed to act as the pathognomonic lesion for TiPV infection as this lesion manifested exclusively in the pancreas and not in other TiPV-infected tissues (Dong et al., 2022). It is worth noting that CAIB are also commonly observed in other parvovirus infections, such as infectious hypodermal and haematopoietic necrosis (IHHNV) in shrimp (Pillai & Bonami, 2012). However, Dong et al. (2022) only examined nine adult Nile tilapia for CAIB manifestation. Hence, future studies should include a larger sample size and different life stages of tilapia to ascertain whether this histopathological marker consistently appears in TiPV-infected fish.

Although the transmission of TiPV remains incompletely understood, evidence suggests potential vertical transmission, as indicated by the detection of TiPV DNA in the oocyte membranes of Nile tilapia ovary (Dong et al., 2022). Given similarities with other aquatic viruses, horizontal transmission of TiPV is also probable (Shinn et al., 2023), aligning with observations in other parvoviruses like hepatopancreatic parvovirus (HPV) in shrimp (Catap et al., 2003). Various diagnostic methods for TiPV exist, including virus isolation and molecular techniques (Dong et al., 2022; Liu et al., 2020; Yamkasem et al., 2022). TiPV has been successfully propagated on tilapia brain cells (TiB) (Liu et al., 2020; Piewbang et al., 2022) and snakehead-fish cell line (E-11) (Piewbang

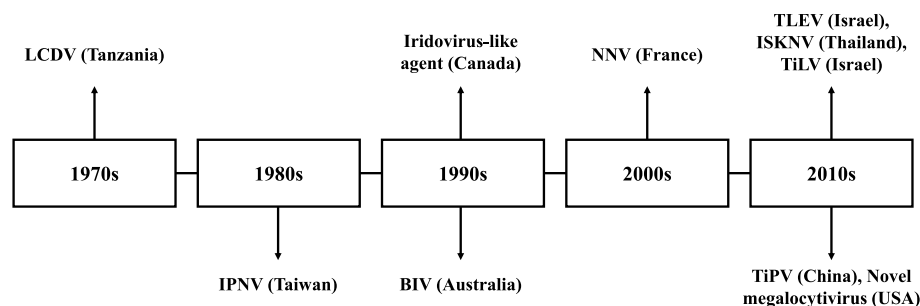


Fig. 1. Timeline of the first reported cases of viral infections and the respective occurrence of incidence in tilapia.

Abbreviations: Bohle iridovirus (BIV); Infectious pancreatic necrosis virus (IPNV); Infectious spleen and kidney necrosis virus (ISKNV); Lymphocystis disease virus (LCDV); Nervous necrosis virus (NNV); Tilapia lake virus (TiLV); Tilapia parvovirus (TiPV); Tilapia larvae encephalitis virus (TLEV).

Table 1

Summary of viral taxonomy, characteristics, tilapia host range, occurrence, target life stages and mortality rates of known viral infections in tilapia.

Pathogen	Taxonomy	Viral characteristics	Tilapia host species	Recorded distribution	Life stages	Mortality rate	References
Tilapia parvovirus (TIPV)	Parvoviridae; <i>Chaphamaparvovirus</i>	ssDNA, non-enveloped, spherical, 28–32 nm	<i>O. niloticus</i> , <i>Oreochromis</i> spp.	China, Thailand	Juvenile, adult	• 60–70% (NI) • 90% (EI)	Liu et al. (2020), Piewbang et al. (2022), Yamkasem, Tattiyapong, et al. (2021)
Tilapia larvae encephalitis virus (TLEV)	Herpesviridae	DNA, icosahedral, electron-dense core, 100 nm	<i>O. aureus</i> , <i>Oreochromis</i> sp., <i>O. niloticus</i> , <i>S. galilaeus</i>	Israel	Larvae	• Above 90% (NI) • 80–95% (EI)	Shlapobersky et al. (2010), Sinyakov et al. (2011)
Infectious spleen and kidney necrosis virus (ISKNV)	Iridoviridae; <i>Megalocyttivirus</i>	dsDNA, enveloped, icosahedral, electron-dense core, 114–160 nm	<i>O. niloticus</i> , <i>O. niloticus</i> × <i>O. mossambicus</i>	Brazil, Ghana, Thailand, USA	Fry, juvenile, adult, broodstock	• 50–90% (NI)	Dong et al. (2015), Figueiredo et al. (2022), Ramírez-Paredes et al. (2021), Subramaniam et al. (2016), Suebsing et al. (2016)
Iridovirus-like agents	Iridoviridae	DNA, icosahedral	<i>O. niloticus</i>	Canada, USA	Fingerlings	Not specified	McGrogan et al. (1998), Smith et al. (1997)
Novel megalocyttivirus	Iridoviridae; <i>Megalocyttivirus</i>	DNA, icosahedral, electron-dense core, 110–121 nm	<i>O. niloticus</i> , <i>Oreochromis</i> sp.	USA	Fingerlings	• 39% (EI)	Shahin et al. (2021)
Bohle iridovirus (BIV)	Iridoviridae; <i>Ranavirus</i>	dsDNA, icosahedral, electron-dense core, 110–140 nm	<i>O. mossambicus</i>	Australia	Fry	• ~100% (NI)	Ariel and Owens (1997), Stöhr et al. (2015)
Lymphocystis disease virus (LCDV)	Iridoviridae; <i>Lymphocystivirus</i>	dsDNA, icosahedral, fibril-like protrusions, 120–340 nm	<i>T. amphi-melas</i> , <i>T. esculenta</i> , <i>T. variabilis</i>	Tanzania	Juvenile	Not specified	Borrego et al. (2017), Paperna (1973)
Nervous necrosis virus (NNV)	Noraviridae; <i>Betanodavirus</i>	ssRNA, non-enveloped, icosahedral, 38–54 nm	<i>O. niloticus</i> , <i>Oreochromis</i> sp.	Egypt, France, Indonesia, Thailand	Larvae, fry, juvenile, adult, broodstock	• 70–100% (NI)	Bigarré et al. (2009), Keawcharoen et al. (2015), Prihartini et al. (2015), Taha et al. (2020), Yanuhar et al. (2017)
Infectious pancreatic necrosis virus (IPNV)	Birnaviridae; <i>Aquabirnavirus</i>	dsRNA, icosahedral, non-enveloped, 60 nm	<i>O. spilurus</i> , <i>O. mossambicus</i> , <i>O. niloticus</i>	Kenya, Taiwan	Not specified	• No mortality (NI) • 25% (EI)	Hedrick et al. (1983), Manguwiryono and Agius (1987), Mulei et al. (2018)
Tilapia lake virus (TiLV)	Amnoonviridae; <i>Tilapinevirus</i>	ssRNA, enveloped, icosahedral, 55–100 nm	<i>S. galilaeus</i> , <i>T. zillii</i> , <i>T. intermedia</i> , <i>O. niloticus</i> , <i>O. aureus</i> , <i>O. mossambicus</i> , <i>Oreochromis</i> sp., <i>O. niloticus</i> × <i>O. mossambicus</i> , <i>O. niloticus</i> × <i>O. aureus</i>	Bangladesh, Colombia, Ecuador, Egypt, India, Indonesia, Israel, Mexico, Malaysia, Peru, Philippines, Taiwan, Tanzania, Thailand, Uganda, USA	Fertilised eggs, larvae, fry, fingerlings, juveniles, adults, broodstocks	• 5–100% (NI) • 56–86% (EI)	Aich et al. (2022), Amal et al. (2018), Basri et al. (2020), Dong, Ataguba, et al. (2017), Dong, Siriroob, et al. (2017), Eyngor et al. (2014), Ferguson et al. (2014)

Viral characteristics abbreviations: Deoxyribonucleic acid (DNA); Double-stranded DNA (dsDNA); Double-stranded RNA (dsRNA); Ribonucleic acid (RNA); Single-stranded DNA (ssDNA); Single-stranded RNA (ssRNA).**Tilapia host species abbreviations:** *Oreochromis aureus* (*O. aureus*); *Oreochromis mossambicus* (*O. mossambicus*); *Oreochromis niloticus* (*O. niloticus*); *Oreochromis spilurus* (*O. spilurus*); *Sarotherodon galilaeus* (*S. galilaeus*); *Tilapia amphi-melas* (*T. amphi-melas*); *Tilapia esculenta* (*T. esculenta*); *Tilapia variabilis* (*T. variabilis*); *Tilapia zillii* (*T. zillii*); *Tristramella simonis intermedia* (*T. intermedia*).**Mortality rate abbreviations:** Natural infection (NI); Experimental infection (EI).

Table 2

Characteristics of the disease induced by the various viral agents and the respective diagnostic methods.

Pathogen	Clinical signs ^a	Histopathology ^a	Cell culture	Molecular diagnostics	References
Tilapia parvovirus (TiPV)	Darting or corkscrew swimming behaviour, lethargy, anorexia, exophthalmia, ocular lesions and body haemorrhages	Infiltration of lymphocytes in the spleen, liver, kidney and brain. Kidneys showed increase in MMCs. Necrosis was evident in the spleen and gills. CAIB observed exclusively in pancreas	TiB, E-11	PCR, multiplex PCR, qPCR, dPCR, LAMP, IFA, ISH	Liu et al. (2020), Phusantisampan et al. (2022), Piewbang et al. (2022), Zhao et al. (2023)
Tilapia larvae encephalitis virus (TLEV)	Whirling syndrome and skin discolouration	Not done	Not done	PCR	Shlapobersky et al. (2010), Sinyakov et al. (2011)
Infectious spleen and kidney necrosis virus (ISKNV)	Abnormal movements, lethargy, skin haemorrhages, gill pallor, abdominal distension and ascites	IBC in the spleen, kidney, intestine, liver and heart. Necrosis of the heart, spleen, liver and kidney tissues	GF, BF-2, E-11	PCR, multiplex PCR, qPCR, blue-LAMP, <i>in situ</i> LAMP, WGS using multiplex tiled PCR	Alathari et al. (2023), Figueiredo et al. (2022), Ramírez-Paredes et al. (2021), Subramaniam et al. (2016), Suebsing et al. (2016)
Iridovirus-like agents	Lethargy, bloating, skin darkening, ocular lesions, severe abdominal ascites and organ pallor	Hypertrophied cells in the liver, spleen and kidney. Necrosis present in affected tissues	Not done	Not done	McGrogan et al. (1998), Smith et al. (1997)
Novel megalocytivirus	Gill pallor, fin degradation, abdominal distension, ascites, spleen and liver enlargement	Necrosis and reduction of cell number in the kidney and spleen tissues	OmB	qPCR	Shahin et al. (2021)
Bohle iridovirus (BIV)	Rapid cork-screw swimming pattern (spinning tilapia syndrome)	Myolysis of the muscles. Kidneys displayed haemorrhages, infiltration of eosinophilic granular cells and degradation of renal tubule	BF-2	Not done	Ariel and Owens (1997)
Lymphocystis disease virus (LCDV)	Nodular lesions consisting of lymphocystis-infected cell clusters on tail fins	Nodules consisting of hypertrophied cells surrounded by thick eosinophilic capsules. Vesicular cell nuclei present whereas cytoplasm showed large vacuoles and basophilic inclusion bodies	Not done	Not done	Paperna (1973)
Nervous necrosis virus (NNV)	Erratic or cork-screw swimming behaviour, floating at the water surface, eye opacity, skin darkening, lethargy and anorexia	Lesions, including vacuolization, in the nervous tissues. Inclusion bodies, perivascular cuffing and gliosis found in nervous tissues	SSN-1	RT-PCR, nested RT-PCR, IHC, IFA	Bigarré et al. (2009), Keawcharoen et al. (2015), Prihartini et al. (2015), Taha et al. (2020)
Infectious pancreatic necrosis virus (IPNV)	Natural infection associated with no clinical signs. Experimental infection resulted in skin haemorrhages, gastroenteritis and liver enlargement	Leukocyte infiltration in the swim bladder, liver and intestine. Increased eosinophilic granule cells in intestine under experimental infection	CHSE-214	RT-PCR, analysis of IPNV polypeptides and RNA using polyacrylamide gel electrophoresis	Hedrick et al. (1983), Mangunwiryo and Agius (1987), Mulei et al. (2018)
Tilapia lake virus (TiLV)	Lethargy, swimming at surface of the water, appetite loss, anaemia, exophthalmia, corneal opacity, abdominal distension, ascites, and skin erosion, haemorrhage and discolouration	Liver showed syncytial cells, eosinophilic inclusion bodies and increased MMCs. Necrosis of kidney and spleen with increased MMCs in the spleen. Haemorrhages, gliosis, perivascular lymphocytic cuffing, congestion and oedema present in the brain.	E-11, OmB, CFF, OnLB, OnLL, TmB	RT-PCR, nested RT-PCR, semi-nested RT-PCR, RT-qPCR, dPCR, RT-LAMP, Real-time RPA, RT-RPA-Cas12a, ISH, IHC, iELISA	Aich et al. (2022), Behera et al. (2018), Dong, Ataguba, et al. (2017), Dong, Siriroob, et al. (2017), Eyngor et al. (2014), Ferguson et al. (2014), Kembou Tsofack et al. (2017), Sukonta et al. (2022), Tattiyapong et al. (2017), Wang et al. (2021)

Histopathology abbreviations: Cowdry type A inclusion bodies (CAIB); Inclusion body-bearing cells (IBC); Melanomacrophage centres (MMCs).**Cell culture abbreviations:** Blue gill fry-2 cell line (BF-2); *Pristolepis fasciatus* caudal fin cell line (CFF); Chinook salmon embryonic cell line (CHSE-214); Snakehead-fish cell line (E-11); Grunt fin cell line (GF-2); Mozambique tilapia brain cell line (OmB); Nile tilapia brain cell line (OnLB); Nile tilapia liver cell line (OnLL); Snakehead-fish cell line (SSN-1); Tilapia brain cell line (TiB); Tilapia bulbous arteriosus cell line (TmB).**Molecular diagnostics abbreviations:** Digital droplet PCR (dPCR); Indirect enzyme-linked immunosorbent assay (iELISA); Indirect immunofluorescence assay (IFA); Immunohistochemistry (IHC); *In situ* hybridisation (ISH); Loop-mediated isothermal amplification (LAMP); Polymerase chain reaction (PCR); Quantitative PCR (qPCR); Ribonucleic acid (RNA); Recombinase polymerase amplification (RPA); Reverse transcription-LAMP (RT-LAMP); Reverse transcription PCR (RT-PCR); Reverse transcription-RPA-Cas12a (RT-RPA-Cas12a); Reverse transcription qPCR (RT-qPCR); Whole genome sequencing (WGS).^a List of clinical signs and histopathology are not intended to be exhaustive, but represent examples of clinical signs and histopathology reported in diseased tilapia.

et al., 2022) where common cytopathic effects (CPEs) observed in TiB include cell shrinkage and rounding six days post-infection (dpi) (Liu et al., 2020). For the E-11 cell line, however, CPEs observed could not be wholly attributed to TiPV as the cells were also concurrently infected with TiLV (Piewbang et al., 2022).

Various molecular-based diagnostics have been developed to detect TiPV, including standard polymerase chain reaction (PCR) (Liu et al., 2020), multiplex PCR (Prasartset & Surachetpong, 2023), quantitative

PCR (qPCR) (Liu et al., 2020; Yamkasem et al., 2022), digital droplet PCR (dPCR) (Zhao et al., 2023), and loop-mediated isothermal amplification (LAMP) (Phusantisampan et al., 2022). These methods utilize primers designed based on highly conserved NS1 and VP1 genes, making them suitable for molecular diagnostic assays (Liu et al., 2020; Reuter et al., 2022). They demonstrate high specificity for TiPV, with no observed cross-reactivity with other pathogens such as TiLV and grass carp reovirus (GCRV) (Phusantisampan et al., 2022; Yamkasem et al.,

Characteristics of various diseases caused by DNA viruses in tilapia

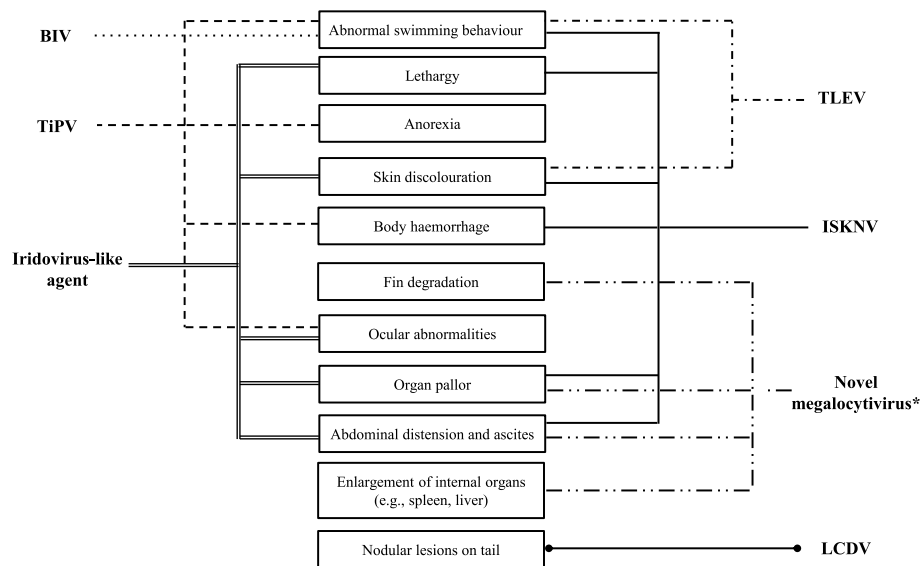


Fig. 2. Disease characteristics exhibited in tilapia after infection with individual DNA viruses.

Note: *Clinical signs observed from novel megalocytivirus infection cannot be conclusively attributed to novel megalocytivirus alone as there was concurrent bacterial infection.

Characteristics of various diseases caused by RNA viruses in tilapia

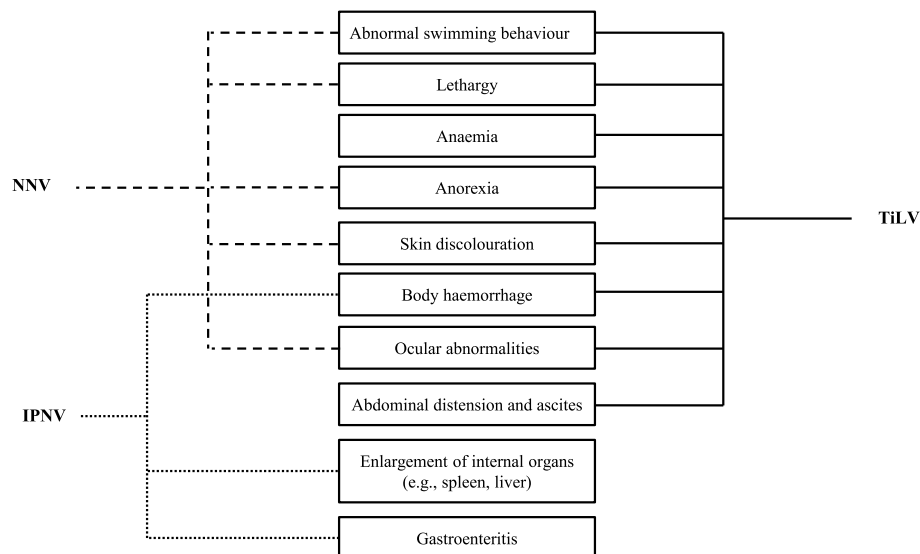


Fig. 3. Disease characteristics exhibited in tilapia after infection with individual RNA viruses.

2022; Zhao et al., 2023). In terms of TiPV detection sensitivity, dPCR showed the lowest detection limit of 0.07 copies/ μ L, compared to 4.63 copies/ μ L in qPCR (Zhao et al., 2023). While qPCR detection assay developed by Yamkasem et al. (2022) demonstrated slightly inferior sensitivity (detection limit of 10 copies/ μ L), it still exhibits roughly 100–1000 times greater sensitivity than conventional PCR. On other hand, LAMP developed by Prasartset and Surachetpong (2023) had a sensitivity of 5.25×10^3 copies/ μ L, suggesting that the assay was not optimised.

Lastly, indirect immunofluorescence assay (IFA) and ISH using anti-TiPV antibody and TiPV-specific probes, respectively, have been used to visualise TiPV localisation in affected samples, yielding positive signals

if TiPV is present (Dong et al., 2022; Liu et al., 2020; Piewbang et al., 2022).

2.2. Tilapia larvae encephalitis virus

Tilapia larvae encephalitis virus (TLEV) is an emerging DNA virus which was first identified in gynogenetic blue tilapia larvae reared in a laboratory in Israel (Shlapobersky et al., 2010). Transmission electron microscopy (TEM) analysis on infected brain tissues revealed 100 nm icosahedral hexagonal virions possessing electron-dense cores (Shlapobersky et al., 2010). Further analysis on the amino acid sequence of a conserved TLEV-based region showed highest levels of conservancy with

other viruses from the Herpesviridae family (Shlapobersky et al., 2010). For instance, pair-wise alignment of TLEV and turtle herpesvirus GTHV-Ha (family: Herpesviridae, subfamily: *Alphaherpesvirinae*) exhibited the greatest similarity (76%) and identity (64%) (Shlapobersky et al., 2010; Yu et al., 2000). These morphological and phylogenetic findings by Shlapobersky et al. (2010) suggest that TLEV may belong to the Herpesviridae family. However, sequence alignment and phylogenetic analysis were carried out based on a 56 amino acid fragment of TLEV DNA polymerase instead of the whole genome sequence. Consequently, the classification of TLEV in Herpesviridae cannot be wholly determined since coding-complete genome sequences are preferred for taxonomic classification of viruses (Aiweksakun & Simmonds, 2018; Shlapobersky et al., 2010).

TLEV was reported to cause natural widespread mortality in blue tilapia larvae where a mortality rate above 90% was recorded (Shlapobersky et al., 2010). High mortality rates were also documented under experimental conditions for both blue and red tilapia (*Oreochromis* sp.) which reached 96% and 80% respectively (Shlapobersky et al., 2010). Similarly, Sinyakov et al. (2011) reported a mortality rate of 90–95% in blue tilapia larvae under experimental conditions. Clinical manifestations in diseased fish were characterised by whirling syndrome, which is a spiral swimming behaviour in both blue and red tilapia larvae (Shlapobersky et al., 2010). Additional clinical signs preceding the onset of whirling syndrome include skin discoloration. Affected blue tilapia developed dark pigmentation, while infected red tilapia exhibited pale skin instead (Shlapobersky et al., 2010; Sinyakov et al., 2011). TLEV was capable of interspecies horizontal transmission where co-habitation of TLEV-infected blue tilapia larvae with other tilapia larvae species such as red tilapia, Nile tilapia and mango tilapia (*Sarotherodon galilaeus*) resulted in TLEV transmission and mortality (Shlapobersky et al., 2010). Vertical transmission of TLEV was also evident as the virus was detected in the offspring of TLEV-positive females and in the reproductive organs of both genders (Shlapobersky et al., 2010; Sinyakov et al., 2011). Shlapobersky et al. (2010) suggested that the virus infected both genders equally, as TLEV was identified in the gametes of 62% males and 56% females. However, TLEV vulnerability may be species dependent, as more blue tilapia (72%) were infected compared to Nile tilapia (33%) (Shlapobersky et al., 2010).

TLEV infection impaired the innate immunity of both horizontally and vertically infected blue and red tilapia, as evidenced by comparable downregulation of reactive oxygen species (ROS) in phagocytes (Sinyakov et al., 2011). This impairment is deleterious for fish larvae, as their adaptive immunity is not yet fully developed (Vadstein et al., 2013). Currently, there is only one PCR-based detection method targeting the TLEV DNA polymerase gene (Shlapobersky et al., 2010). There is no established TLEV isolation method, which impedes additional characterisation of TLEV, such as host-virus interactions, CPEs, burst size and more, all of which are important to obtain a comprehensive view of TLEV impact in tilapia (Trubl et al., 2020). Likewise, histopathological findings in TLEV-infected tilapia remain undetermined, which hinders complete characterisation of TLEV pathology as well as impeding diagnosis of TLEV (Gupta et al., 2009).

While studies on TLEV have been focusing on laboratory fish specimens, TLEV is likely to have existed outside of the laboratory, as Shlapobersky et al. (2010) have reported that subclinical red tilapia, Nile tilapia and mango tilapia adults obtained from external sources were TLEV-positive. However, since no epidemiological studies were conducted, the source and distribution of this virus remain unknown.

2.3. Infectious spleen and kidney necrosis virus

ISKNV is a double-stranded DNA (dsDNA) virus classified within the family Iridoviridae, specifically under the genus *Megalocytivirus* (Chinchar et al., 2005). ISKNV represents one of the three genotypes among megalocytiviruses, alongside red sea bream iridovirus (RSIV) and turbot reddish body iridovirus (TRBIV). The virion structure of ISKNV

comprises enveloped icosahedral particles, ranging approximately 115–200 nm in diameter, with a central electron-dense core (Bermúdez et al., 2018; Fu et al., 2011; Fusianto et al., 2023; Subramaniam et al., 2012). The virus exhibits a broad host range, affecting various freshwater and marine fish species, including mandarin fish (He et al., 2002), grouper (Razak et al., 2014), and gourami (Sukenda et al., 2020).

The first case in tilapia was reported during disease outbreaks in intensive cage cultures of Nile tilapia in Thailand (Dong et al., 2015). The iridovirus in this case was suggested to act as an opportunistic pathogen during this outbreak (Dong et al., 2015) and was thought to be clustered within the ISKNV genotype as confirmed by semi-nested PCR and sequencing analysis of amplicons (GenBank JQ253374.1) (Razak et al., 2014). Additionally, the first comprehensive characterization of ISKNV in tilapia was published by Subramaniam et al. (2016) following an epizootic outbreak among cultivated Nile tilapia fry in the United States, which resulted in high fatality rates (50%–75%). Recent ISKNV outbreaks have been reported in Ghana (Ramírez-Paredes et al., 2021) and Brazil (Figueiredo et al., 2022), with the former demonstrating elevated mortality rates (60%–90%) among Nile tilapia.

Clinical signs such as lethargy, skin darkening, abnormal swimming pattern, skin haemorrhages, gill pallor and distension of the abdomen with associated ascites were manifested in diseased tilapia (Figueiredo et al., 2022; Ramírez-Paredes et al., 2021; Subramaniam et al., 2016). There is evidence of vertical transmission of ISKNV in tilapia where *in situ* LAMP on red tilapia (*O. niloticus* × *O. mossambicus*) broodstock gonads revealed ISKNV-positive signals in the ovarian follicles (Suebsing et al., 2016). Likewise, *in vitro* fertilisation of ten red tilapia breeding pairs revealed that two breeding pairs transmitted the virus to the next generation as ISKNV was detected by LAMP in 20% of the fertilised eggs (Suebsing et al., 2016). Furthermore, horizontal transmission of ISKNV in tilapia is plausible, as most offspring from the ten breeding pairs were tested positive for ISKNV in later developmental stages (Suebsing et al., 2016). Interestingly, none of the sampled broodstock in this study exhibited clinical signs, suggesting asymptomatic carriage of ISKNV.

The histopathological presence of inclusion body-bearing cells (IBC) in tissues infected with ISKNV is a distinctive indicator of megalocytivirus infections (Xu et al., 2008). IBCs are characterised by their enlarged, hypertrophied morphology, with basophilic cytoplasm and a centrally or marginally positioned nucleus (Sudthongkong et al., 2002; Volpe, Errani, Mandrioli, & Ciulli, 2023). Recent cases have reported the presence of IBCs in various organs, including the spleen, kidney, intestine, liver, and heart (Figueiredo et al., 2022; Ramírez-Paredes et al., 2021; Subramaniam et al., 2016). However, misdiagnosis can occur due to the potential for IBCs to be mistaken for amoebae, and their presence has also been observed in non-megalocytivirus cases (Go et al., 2016). Furthermore, histopathology alone cannot determine the megalocytivirus genotype, necessitating the use of molecular diagnostics to distinguish between different genotypes and confirm ISKNV infections (Go et al., 2016; Volpe, Errani, Mandrioli, & Ciulli, 2023).

At present, several molecular diagnostics have been developed for ISKNV detection in tilapia which include PCR (Subramaniam et al., 2016; Suebsing et al., 2016), qPCR (Ramírez-Paredes et al., 2021), multiplex PCR (Prasartset & Surachetpong, 2023), and LAMP (Suebsing et al., 2016). Generally, LAMP showed high levels of sensitivity where the detection limit was 1 copy of ISKNV, which was 100 times more sensitive than PCR (Suebsing et al., 2016). Similarly, qPCR could also detect up to 1 copy of ISKNV in infected tilapia (Ramírez-Paredes et al., 2021). While LAMP is advantageous for point-of-care diagnostics, it does not provide quantitation of the viral load as in qPCR.

Recently, Alathari et al. (2023) developed a novel multiplex tiled PCR method for rapid whole-genome sequencing (WGS) of ISKNV in Nile tilapia, which has the potential to significantly enhance our understanding of the epidemiology of viral diseases and inform disease control strategies. Furthermore, viral isolation from infected tilapia has been successful on several cell lines, including the grunt fin cell line (GF), blue gill fry-2 cell line (BF-2), and E-11, with CPEs such as cell

enlargement observed mainly in the BF-2 cell line, while the GF cell line showed no CPEs (Ramírez-Paredes et al., 2021).

2.4. Iridovirus-like agent

Two previous studies have documented iridovirus-like infections in tilapia in the USA (Smith et al., 1997) and Canada (McGrogan et al., 1998). Notably, the affected Nile tilapia fingerlings in both studies were imported from the same region, specifically Florida, USA (McGrogan et al., 1998; Smith et al., 1997). The ensuing infections in both cases exhibited similar symptoms, including lethargy, abdominal bloating, skin darkening, ocular lesions, severe abdominal ascites, and pallor of the internal organs (McGrogan et al., 1998; Smith et al., 1997). TEM examination revealed multiple polyhedral virions with structures indicative of an iridovirus, while histopathological analysis showed hypertrophied cells, cellular necrosis, and necrotic cellular and nuclear debris in affected tissues in both studies (McGrogan et al., 1998; Smith et al., 1997).

Recently, Subramaniam et al. (2016) suggested that these iridovirus-like infections might be isolates of ISKNV, as BLASTN analysis of the major capsid protein (MCP) gene revealed a 100% identity between the MCP gene sequences from Smith et al. (1997) and Subramaniam et al. (2016), with another reference ISKNV gene sequence (GenBank NC_003494). Considering that the case by McGrogan et al. (1998) shares many similarities with Smith et al. (1997) and Subramaniam et al. (2016), the aetiological agent involved is likely ISKNV. However, molecular testing on the MCP gene sequence should be conducted on archived samples from the by McGrogan et al. (1998) outbreak for complete validation.

2.5. Novel megalocytivirus

Recently, a novel megalocytivirus was identified and characterised in tilapia fingerlings (*Oreochromis* spp.) in an aquaculture farm in the USA, following an epizootic outbreak with concurrent bacterial infection (*Francisella orientalis*) (Shahin et al., 2021). The novel megalocytivirus was isolated using the Mozambique tilapia brain cell line (OmB), where subsequent TEM analysis revealed an electron-dense virion with an icosahedral nucleocapsid, measuring 110–121 nm in diameter (Shahin et al., 2021), consistent with the characteristics of megalocytiviruses described in previous literature (Kurita & Nakajima, 2012). In OmB cells, the virus induced CPEs including cell rounding, vacuolation, and detachment from the monolayer within 6–10 days post-infection (Shahin et al., 2021). Phylogenetic analysis of the MCP and DNA polymerase genes revealed a 92% similarity with scale drop disease virus (SDDV), suggesting a close relationship between the two viruses (Shahin et al., 2021). Further phylogenetic analysis of the MCP and DNA polymerase amino acid sequences supported the classification of the novel megalocytivirus within the genus *Megalocytivirus*, where it formed a distinct branch closely related to SDDV and European chub iridovirus (ECIV) (Shahin et al., 2021).

Several clinical manifestations have been observed in naturally diseased fish, including abdominal distension, ascites, gill pallor, fin erosion, and spleen and liver enlargement (Shahin et al., 2021). However, it is challenging to determine whether these clinical signs were primarily caused by the novel megalocytivirus, as a concurrent bacterial infection was present in diseased fish, and autolytic changes were reported in the moribund tilapia (Shahin et al., 2021), which can impede the accuracy of gross lesion interpretation (George et al., 2016). Histopathological analysis revealed necrosis and a reduction in cell number in the kidney and spleen tissues of experimentally IP-injected Nile tilapia fingerlings. Additionally, a mortality rate of 39% was recorded in experimentally infected Nile tilapia (Shahin et al., 2021). These clinical signs and histopathology differ from those typically associated with SDDV (Gibson-Kueh et al., 2012; Nurliyana et al., 2020), suggesting that the virus in question is novel.

Shahin et al. (2021) designed a TaqMan qPCR assay to detect an 80 bp region of the MCP gene of this novel megalocytivirus, with a detection limit of 44 copies/reaction. However, further tests should be conducted using this diagnostic assay to determine if there is any cross-reactivity with other viruses, especially other similar but genetically different megalocytiviruses (Kralik & Ricchi, 2017).

2.6. Bohle iridovirus

BIV is a member of the family Iridoviridae and genus *Ranavirus*, which are known for their broad host range across multiple animal classes, such as amphibians (Cullen & Owens, 2002), reptiles (MacLaine et al., 2019), turtles (Wirth et al., 2019), and fish (Moody & Owens, 1994). Ranaviruses are large dsDNA viruses with icosahedral symmetry, measuring 120–180 nm in diameter (Stöhr et al., 2015). Notably, a study by Ariel and Owens (1997) remains the only documented case of BIV infection in tilapia, specifically in Mozambique tilapia fry, in an Australian laboratory, where mortality rates approached 100%. Infected tilapia exhibited abnormal swimming behaviour, characterised by a fast, corkscrew-like pattern, which researchers termed "spinning tilapia (ST) syndrome." Additionally, moribund tilapia displayed various histopathological lesions, including myolysis in the muscles, haemorrhages, renal tubule degeneration, and eosinophilic granular cell infiltration in the kidneys (Ariel & Owens, 1997).

Despite the unsuccessful viral isolation from infected Mozambique tilapia onto the BF-2 cell line, a barramundi bioassay was employed to validate whether BIV was the causative agent, given barramundi's high susceptibility to BIV (Ariel & Owens, 1997; Moody & Owens, 1994). Healthy barramundi (*Lates calcarifer*) fingerlings were fed infected tilapia that resulted in an infection that caused erratic swimming patterns and uncontrolled movement in the fish, accompanied by necrosis in the kidney and spleen (Ariel & Owens, 1997; Moody & Owens, 1994). Furthermore, uninfected tilapia also developed ST syndrome after consuming diseased tilapia (Ariel & Owens, 1997), providing evidence of horizontal transmission of the virus through the oral route. Additionally, there is evidence of vertical transmission, as females that survived the initial mortality episode produced offspring exhibiting abnormal swimming behaviour with recurring mortality episodes (Ariel & Owens, 1997).

TEM examination of the kidneys of moribund barramundi revealed icosahedral viral particles measuring 110–140 nm in diameter with an electron-dense core, consistent with the typical morphology of iridoviruses (Ariel & Owens, 1997). This finding, combined with the results of the barramundi bioassay and subsequent histopathological analysis, confirmed BIV as the aetiological agent. Notably, attempts to isolate the virus from infected barramundi onto the BF-2 cell line were unsuccessful in this study, contrasting with previous studies that utilized the same method and cell line (Moody & Owens, 1994). This suggests that the inability to propagate the virus may not be due to the use of an incompatible cell line, but rather to other factors (Felten et al., 2022). One possible factor could be the change in viral infectivity within the infected tilapia, which is supported by another incident where BIV isolation was unsuccessful after passage through tilapia (Ariel & Owens, 1997). Further research, however, is needed to validate this hypothesis.

Currently, there is a lack of molecular diagnostics for BIV in tilapia, despite these diagnostics having been primarily established for non-piscine organisms like reptiles, amphibians, and turtles (Ariel et al., 2015; MacLaine et al., 2019; Wirth et al., 2019). Therefore, it is essential to test the application of existing molecular diagnostics in BIV-infected tilapia to determine their suitability for BIV diagnostics in this species.

2.7. Lymphocystis disease virus

LCDV, a member of the family Iridoviridae and genus *Lymphocystis*, is a double-stranded DNA (dsDNA) virus characterised by an icosahedral capsid with fibril-like external protrusions arranged as a

fringe (Borrego et al., 2017; Hossain et al., 2008). The virion diameter ranges from 120 to 340 nm, varying depending on the host species (Borrego et al., 2017; Paperna et al., 2001). LCDV is the causative agent of lymphocystis disease, a chronic condition affecting freshwater and marine fishes worldwide (Borrego et al., 2017). The disease is characterised by the formation of pearl or wart-like nodules, typically seen on the fins, skin, or gills, although other tissues may be affected (Borrego et al., 2017; Kopper et al., 2014; Xu et al., 2014).

The only recorded LCDV infection in tilapia was by Paperna (1973), which affected juvenile tilapia species including Manyara tilapia (*O. amphimelas*), Singida tilapia (*O. esculenta*), and Victoria tilapia (*O. variabilis*) in Tanzania, Africa. As a result of LCDV infection, nodular lesions consisting of clusters of lymphocystis-infected cells were documented on the tails of juvenile tilapia (Paperna, 1973), a characteristic of LCDV infection (Borrego et al., 2017; Huang et al., 2015). Histopathological examination of the nodules revealed rounded hypertrophied cells encapsulated by thick eosinophilic capsules with large vacuoles and basophilic inclusion bodies present in the cytoplasm (Paperna, 1973). Correspondingly, the nuclei of the cells appeared to be vesicular and were found to be in various stages of disintegration.

According to Leiva-Rebollo et al. (2020), there is evidence of horizontal and vertical transmission of LCDV in other fish species, although this remains to be validated in tilapia. To date, histopathological diagnostics remain the only published method to detect LCDV infection in tilapia. However, other molecular diagnostics have been developed for different fish species, including PCR (Kvitt et al., 2008), qPCR (Ciulli et al., 2015), LAMP (Li et al., 2010), immunohistochemistry (IHC), and ISH (Cano et al., 2009). Viral isolation of LCDV from infected grouper onto cell lines such as BF-2 and *Sparus aurata* fibroblast (SAF-1) has also been established (Huang et al., 2015). Therefore, the application of these detection and isolation methods could also be applicable for LCDV in tilapia.

2.8. Nervous necrosis virus

NNV, the aetiological agent of viral nervous necrosis (VNN) disease or viral encephalopathy and retinopathy (VER) (Yang et al., 2022), belongs to the family Nodaviridae under the genus *Betanodavirus*. Four genotypes are formally recognized by the International Committee on Taxonomy of Viruses (ICTV, 2022): barfin flounder NNV (BFNNV), red-spotted grouper NNV (RGNNV), striped jack NNV (SJNNV), and tiger puffer NNV (TPNNV). Additionally, other genotypes, such as turbot nodavirus (TNV) (Johansen et al., 2004) and Korean shellfish NNV (KSNNV) (Kim et al., 2019), have been identified but are yet to be formally classified by ICTV. NNV has a broad host range, affecting over 120 wild and farmed fish species, with RGNNV being the most prevalent due to its ability to infect the largest number of host species (Bandín & Souto, 2020; Jungi et al., 2022). Structurally, NNV is a non-enveloped, single-stranded RNA (ssRNA) virus, approximately 25–34 nm in diameter, with an icosahedral capsid (Bandín & Souto, 2020; Shetty et al., 2012). The viral genome consists of two RNA segments: RNA1 and RNA2, with phylogenetic analysis distinguishing between genotypes based on the T4 domain within the latter segment (Shetty et al., 2012).

NNV outbreaks in tilapia have been reported in several instances, occurring in France (Bigarré et al., 2009), Thailand (Keawcharoen et al., 2015), Indonesia (Prihartini et al., 2015; Yanuher et al., 2017), and most recently in Egypt (Taha et al., 2020), with mortality up to 90%–100% in farmed Nile tilapia larvae and 70% in fry (Keawcharoen et al., 2015; Taha et al., 2020). Phylogenetic analysis of NNVs isolated from these outbreaks were clustered within the RGNNV genotype with the sequences from Keawcharoen et al. (2015) and Taha et al. (2020) showing high homology (93%–99%) when compared with deposited RGNNV sequences. Interestingly, TEM analysis showed that the icosahedral virions of these NNVs were larger (38–54 nm in diameter) (Taha et al., 2020) than standard NNVs. The variation of size could be due to mutations in the NNV genome that affects the capsomere number and

structure (Taha et al., 2020), suggesting the Egyptian strain could be novel.

NNV infection in tilapia manifests with several clinical signs, including abnormal swimming behaviour such as floating at the water surface and erratic or corkscrew-like swimming patterns (Bigarré et al., 2009; Keawcharoen et al., 2015; Taha et al., 2020; Yanuher et al., 2018). Other signs include eye opacity, darkening of the fish body, lethargy, and anorexia (Bigarré et al., 2009; Taha et al., 2020; Yanuher et al., 2018). Notably, in some tilapia samples, NNV infection did not result in major clinical signs, despite detection via molecular diagnostics (Prihartini et al., 2015; Yanuher et al., 2017), consistent with previous studies indicating several subclinical fish species were NNV-positive (Gomez et al., 2006; Naveen Kumar et al., 2017). Consequently, this suggests that asymptomatic tilapia may act as carriers for NNV (Prihartini et al., 2015).

According to Bandín and Souto (2020), NNV is capable of both vertical and horizontal transmission. In tilapia, evidence suggests vertical transmission, with NNV found in broodstock ovaries (Taha et al., 2020), and supported by Bigarré et al. (2009), who observed no NNV outbreaks post egg sterilization. Horizontal transmission is implicated in the French outbreak, with imported fish potentially introducing and spreading NNV (Bigarré et al., 2009). However, further research on NNV transmission in tilapia should be conducted to obtain direct evidence of both vertical and horizontal transmissions.

Histopathological examination of deceased tilapia revealed lesions in the neurological tissues (e.g., brain, retina, spinal cord, and olfactory bulb) that displayed vacuolization (Bigarré et al., 2009; Keawcharoen et al., 2015), a classical sign of NNV infection (Shetty et al., 2012). Other histopathological changes in the neurological tissues include the presence of inclusion bodies (Prihartini et al., 2015), perivascular cuffing, and gliosis (Keawcharoen et al., 2015).

Clinical signs and histopathological diagnosis alone may not be reliable for identifying NNV infection in tilapia due to the overlapping symptoms it shares with TiLV (Bigarré et al., 2009; Eyngor et al., 2014; Keawcharoen et al., 2015). This highlights the necessity of employing molecular methods to distinguish between the two viruses in regions where NNV and TiLV are endemic. As such, reverse transcription-PCR (RT-PCR) (Bigarré et al., 2009; Keawcharoen et al., 2015; Taha et al., 2020) and nested RT-PCR (Prihartini et al., 2015; Yanuher et al., 2017) targeting the capsid gene and T4 domain of NNV were utilized to validate the presence of the virus in tilapia.

Furthermore, IFA which involves virus isolation using the snakehead-fish cell line (SSN-1) was also used to detect NNV in tilapia, as demonstrated by Bigarré et al. (2009). Despite the absence of CPES, fluorescent signals indicative of NNV presence were observed. Similarly, Keawcharoen et al. (2015) employed IHC using a polyclonal antibody specific to NNV to detect positive signals in neurological tissues from moribund tilapia. However, the sensitivity and specificity of these assays will be affected greatly by the quality of the antibodies used.

2.9. Infectious pancreatic necrosis virus

IPNV belongs to the family Birnaviridae and the genus *Aquabirnavirus*. This non-enveloped virus is characterised by a bisegmented (segment A and B) double-stranded RNA (dsRNA) genome, enclosed within an icosahedral capsid measuring approximately 60 nm in diameter (Oidtmann et al., 2018; Tapia et al., 2022). IPNV is widely distributed and infects various fish species globally (Dopazo, 2020). Initially, aquabirnaviruses were classified based on three reference serotypes (VR-299, Sp, and Ab) determined through anti-serum neutralization tests (Tapia et al., 2022). However, the classification system evolved, dividing aquabirnaviruses into two serogroups (A and B), encompassing nine and one serotypes, respectively (Hill & Way, 1995). Presently, a molecular classification system identifies seven genogroups based on variations in segment A (Blake et al., 2001; Nishizawa et al., 2005), with a correlation observed between type strains, serotypes, and

genogroups.

To date, the number of reported cases of IPNV infection in tilapia remains limited. Published occurrences of IPNV infections include those documented in Taiwan (Hedrick et al., 1983) and Kenya (Mulei et al., 2018). Through neutralization tests and comparisons of IPNV polypeptides and RNA with the three original serotypes, Hedrick et al. (1983) determined that the Taiwan isolated strain from tilapia was closely related to the European Ab serogroup (now classified as serotype A3 and belongs to genogroup 2). The Kenyan isolate was found to be identical to the European Sp isolate based on phylogenetic analysis of IPNV segment A (Mulei et al., 2018), (serotype A2 belonging to genogroup 5) (Dopazo, 2020; Tapia et al., 2022).

While no mortality rates were recorded in these instances, Mangunwiryo and Agius (1987) reported a 25% fatality rate among Sabaki tilapia (*O. spilurus*) after experimental IPNV infection via intramuscular (IM) injection. Although horizontal and vertical transmission of IPNV has been well-documented in other fish species (Bootland et al., 1991; Dopazo, 2020), evidence regarding transmission routes in tilapia remains scant. Horizontal transmission of IPNV in tilapia has been suggested; for instance, in the Kenyan case, effluent water from IPNV-affected rainbow trout farms was discharged into the same water source utilized by tilapia farms (Mulei et al., 2018). Consequently, there was a potential for horizontal transmission through contaminated water. However, this hypothesis requires further validation, as the water was not tested for the presence of IPNV.

Clinical signs associated with natural IPNV infection in tilapia remain uncertain as no clinical signs were observed in the Kenyan case (Mulei et al., 2018). Meanwhile, the Taiwan study observed that the clinical signs in diseased Mozambique tilapia could be due to concurrent infections with *Staphylococcus* sp. and *Streptococcus* sp. instead of IPNV (Hedrick et al., 1983). The absence of clinical signs despite IPNV detection aligns with previous findings in various fish species (Mutoloki et al., 2016). However, in experimentally infected Sabaki tilapia, ventrally-located skin haemorrhages, gastroenteritis, and hepatomegaly were observed (Mangunwiryo & Agius, 1987). Histopathological examination of experimentally infected tilapia revealed notable changes, including leukocyte infiltration in the swim bladder, liver, and intestine. Additionally, the intestine exhibited an increased presence of eosinophilic granule cells (Mangunwiryo & Agius, 1987).

Currently, IPNV diagnostics in tilapia are limited, including viral propagation (Hedrick et al., 1983) and two-step RT-PCR targeting IPNV segment A (Mulei et al., 2018). Viral propagation involved the use of Chinook salmon embryonic cell line (CHSE-214), where significant CPEs were observed after three days; however, specific details regarding the types of CPEs were not provided.

2.10. Tilapia lake virus

In 2009, Israel experienced mass mortality events among wild and farmed tilapia, which were attributed to the outbreak of TiLV (Eyngor et al., 2014). Subsequent research has revealed the global prevalence of TiLV, with reported cases in Asia, America, and Africa, as detailed in Table 1 (Ahasan et al., 2020; Contreras et al., 2021; Mugimba et al., 2018; Nicholson et al., 2017). Consequently, TiLV has been recognized by the World Organisation for Animal Health (WOAH) as an emerging disease affecting finfish, prompting discussions regarding its inclusion in the list of reportable finfish diseases (Sood et al., 2021). While initially classified as an orthomyxo-like virus (Bacharach et al., 2016), recent taxonomic revisions have placed TiLV within the family Amnoonviridae, genus *Tilapinevirus*, and species *Tilapia tilapinevirus* (ICTV, 2021). TiLV is characterised as an enveloped, icosahedral single-stranded RNA virus, with diameters ranging from 55 nm to 100 nm (Dong, Siriroob, et al., 2017; Eyngor et al., 2014). Its genome comprises ten segments, each containing an open reading frame (ORF) potentially encoding for 14 distinct proteins (Acharya et al., 2019).

To date, TiLV infects various farmed and wild tilapia species,

including Nile tilapia (Ferguson et al., 2014), red hybrid tilapia (*O. niloticus* × *O. mossambicus*) (Amal et al., 2018), mango tilapia (Eyngor et al., 2014), and others as listed in Table 1. While TiLV has been traditionally regarded as specific to tilapines, evidenced by the absence of TiLV detection in co-cultivated species like milkfish, grey mullet, pearl spot, and common carp during TiLV outbreaks in tilapia (Behera et al., 2018; Eyngor et al., 2014; Fathi et al., 2017), recent reports challenge this notion, suggesting a broader host range for TiLV. For instance, giant gourami (Chiamkunakorn et al., 2019; Jaemwimol et al., 2018), tinfoil barb (Abdullah et al., 2018, 2022), barramundi (Piamsomboon & Wongtavatchai, 2021), and African cichlid (Yamkasem, Piewbang, et al., 2021) have been identified as susceptible to TiLV infection. This expanding host spectrum underscores the necessity for further investigations into the host range of TiLV.

TiLV infection in tilapia has also been associated with secondary infections caused by various pathogens, including *Aeromonas* spp., *Flavobacterium* spp., *Lactococcus garvieae*, *Streptococcus* spp., and TiPV (Nicholson et al., 2017; Surachetpong et al., 2017; Swaminathan et al., 2021; Yamkasem, Tattiyapong, et al., 2021). There is also evidence that synergistic effects may result from co-infection, where Nicholson et al. (2020) demonstrated that tilapia simultaneously infected with TiLV and *Aeromonas hydrophila* yielded greater mortalities compared to single TiLV or *A. hydrophila* infection under experimental conditions. According to Roy et al. (2021), younger red tilapia (*Oreochromis* spp.) (~8 weeks old; 5 g) were more vulnerable to TiLV infection compared to older red tilapia (~12–16 weeks; 25–65 g) as the former had a higher mortality rate compared to the latter during an experimental challenge. This is supported by Yamkasem et al. (2019), who found that the total mortality of farmed fry was higher at 90%–100% compared to broodstock at 5%–10%. Nevertheless, all life stages of tilapia are at risk of contracting TiLV, including broodstocks, adults, juveniles, fingerlings, fry, yolk-sac larvae, and fertilised eggs (Basri et al., 2020; Debnath et al., 2020; Dong, Ataguba, et al., 2017; Pulido et al., 2019).

Furthermore, the mortality rates of TiLV-infected tilapia exhibit significant variation across distinct regions, likely attributed to differences in tilapia species or strains, farming conditions, geographical locations, and other abiotic or biotic factors (Aich et al., 2022; Sunarto et al., 2022). For instance, severe mortality rates in cultivated tilapia have been documented in Thailand (20%–90%), Ecuador (10%–80%), and India (>85%) (Behera et al., 2018; Ferguson et al., 2014; Surachetpong et al., 2017). Conversely, low mortality rates have been reported in Egypt (5%–15%), Taiwan (6%), and Malaysia (15%) (Fathi et al., 2017; WOA, 2017a, 2017b). Additionally, tilapia experimentally challenged with TiLV also displayed high levels of mortality, where intraperitoneal injection of TiLV resulted in 75%–86% mortality in Nile tilapia and 66% mortality in red tilapia (*Oreochromis* spp.) (Eyngor et al., 2014; Tattiyapong et al., 2017), likely attributable to the higher viral load administered to the experimental fish.

Both horizontal and vertical transmission routes of TiLV in tilapia have been elucidated by past studies. Yamkasem et al. (2019) and Dong et al. (2020) suggested that TiLV could be transferred vertically to offsprings, as TiLV was detected in the gonads and eggs of red hybrid tilapia (*Oreochromis* spp.) and Nile tilapia broodstock following an experimental challenge. Experiments involving cohabitation of TiLV-infected and healthy tilapia resulted in healthy tilapia acquiring TiLV, with mortalities varying from 56% to 80%, thus confirming horizontal transmission of TiLV (Eyngor et al., 2014; Liamnimitr et al., 2018). Besides cohabitation, other potential sources of horizontal transmission in tilapia have been identified. For instance, TiLV has been detected in tilapia mucus, faeces, and water used to house TiLV-infected fish (Liamnimitr et al., 2018; Pierezan et al., 2019), suggesting that exposure to these sources may induce TiLV infection in healthy fish. Subsequently, since tilapia species are naturally territorial, aggressive, and have displayed cannibalism (Gonçalves-de-Freitas et al., 2019; Soto et al., 2016), horizontal transmission of TiLV through the intragastric route is possible through the consumption of infected moribund tilapia.

TiLV-afflicted tilapia exhibit numerous clinical signs with variations depending on the locations of TiLV outbreaks. Clinical signs include skin abrasions, skin haemorrhages, discoloration, and ocular lesions (e.g., exophthalmia and corneal opacity) (Behera et al., 2018; Eyngor et al., 2014; He et al., 2023; Tattiyapong et al., 2017). Anaemia, abdominal distension, ascites, protrusion of scales, and scale loss are also other clinical signs induced by TiLV (Behera et al., 2018; del-Pozo et al., 2017; Surachetpong et al., 2017; Turner et al., 2023). Abnormal behaviour from TiLV-affected tilapia has been documented, including lethargy, appetite loss, and swimming at the water's surface (Dong, Ataguba, et al., 2017; Surachetpong et al., 2017). Notably, asymptomatic infection with TiLV was also reported by Senapin et al. (2018), where low viral loads were found in tilapia tissues.

Multiple histopathological lesions in various organs are associated with TiLV infection, where the formation of syncytial cells in the liver is typically observed in naturally and experimentally diseased tilapia (Debnath et al., 2020; Jaemwimol et al., 2018; Turner et al., 2023). Additional histopathological signs in the liver include the presence of eosinophilic inclusion bodies within the liver cells and increased melanomacrophage centres (MMCs) (Debnath et al., 2020; Eyngor et al., 2014; Ferguson et al., 2014). These histopathological findings, along with necrosis, have also been observed in the spleen (Eyngor et al., 2014; Tattiyapong et al., 2017). Meanwhile, histopathological findings in the brain revealed congestion, oedema, haemorrhages, gliosis, and perivascular lymphocytic cuffing (Eyngor et al., 2014; Fathi et al., 2017; Tattiyapong et al., 2017), with some studies reporting the presence of syncytial cells (Behera et al., 2018; Debnath et al., 2020). Eyngor et al. (2014) also reported ocular lesions characterised by ruptured lenticular capsules and the formation of cataracts within the lens. Furthermore, necrosis in the kidney (Fathi et al., 2017; Tattiyapong et al., 2017) and gastric glands (del-Pozo et al., 2017; Ferguson et al., 2014) has been identified, with congestion and haemorrhages also present in the former (Fathi et al., 2017; Turner et al., 2023).

To date, numerous molecular methods have been developed to detect TiLV, including various PCR methods such as RT-PCR (Eyngor et al., 2014; Nicholson et al., 2018), nested RT-PCR (Kembou Tsoufack et al., 2017; Piamsomboon & Wongtavatchai, 2021), semi-nested RT-PCR (Dong, Siriroob, et al., 2017; Taengphu et al., 2020), TaqMan RT-qPCR (Megarani et al., 2022; Waiyemitra et al., 2018), SYBR green RT-qPCR (Chengula et al., 2022; Tattiyapong et al., 2018), and dPCR (Shahi et al., 2023). Other molecular methods include RT-LAMP (Kampeera et al., 2021; Phusantisampan et al., 2019; Yin et al., 2019), real-time recombinase polymerase amplification (real-time RPA) (Wang et al., 2021), reverse transcriptase-RPA-Cas12a (RT-RPA-Cas12a) (Sukonta et al., 2022), ISH (Dong, Siriroob, et al., 2017), IHC (Piewbang et al., 2021), and indirect enzyme-linked immunosorbent assay (iELISA) (Hu et al., 2020; Valsalam et al., 2023).

Among the PCR-based assays, dPCR typically exhibits the highest sensitivity, followed by RT-qPCR, nested RT-PCR, and RT-PCR (Dong et al., 2023; Li et al., 2022; Smith & Osborn, 2009). For instance, Shahi et al. (2023) demonstrated that the detection limit of TiLV using dPCR and RT-qPCR was 3.3 copies/reaction and 29.97 copies/reaction, respectively, indicating approximately ten-fold greater sensitivity than RT-qPCR. Nevertheless, RT-qPCR remains a sensitive detection method, with the TiLV detection limit of SYBR green RT-qPCR being 2 copies/ μ L, which is a hundred times more sensitive than RT-PCR (Tattiyapong et al., 2018). Similarly, Megarani et al. (2022) reported that TaqMan RT-qPCR had a detection limit of 10 TiLV copies/reaction. Nested RT-PCR and semi-nested RT-PCR could achieve comparable sensitivity to TaqMan RT-qPCR, with detection limits of 7 TiLV copies/reaction (Kembou Tsoufack et al., 2017) and 7.5 TiLV copies/reaction (Dong, Siriroob, et al., 2017), respectively. However, direct comparison of detection limits between these assays is challenging due to differences in experimental parameters, such as the amplification of different TiLV segments and the use of different primer sets.

RT-RPA-Cas12a had a detection limit of 200 copies/reaction

(Sukonta et al., 2022), while real-time RPA documented a detection limit of 620.29 copies/ μ L, which was ten times less sensitive than RT-qPCR (62.029 copies/ μ L) (Wang et al., 2021). Despite these assays not being as sensitive as RT-qPCR, an advantage of real-time RPA and RT-RPA-Cas12a is the shorter reaction time, allowing for faster disease diagnosis (Dong et al., 2023; Sukonta et al., 2022; Wang et al., 2021). Additionally, RT-RPA-Cas12a holds the potential for on-site diagnostics since lateral flow detection is available (Sukonta et al., 2022).

On other hand, RT-LAMP demonstrated comparable sensitivity to dPCR, with a TiLV detection limit of 1–1.6 copies/reaction (Kampeera et al., 2021; Yin et al., 2019). RT-LAMP employs multiple primers, which may lead to false-positive results due to increased chances of inaccurate primer binding and the formation of primer dimers (Dong et al., 2023; Sukonta et al., 2022). However, upon careful optimisation and validation, RT-LAMP could become a powerful tool for TiLV diagnostics due to its superior sensitivity and point-of-care capability.

Amidst these molecular diagnostics, PCR techniques remain the most frequently used for TiLV detection and diagnosis (Megarani et al., 2022). WOAHA has recognized four PCR assays for the detection and diagnosis of TiLV detection and diagnosis (WOAHA, 2022), with two assays having been previously described (Dong, Siriroob, et al., 2017; Tattiyapong et al., 2018), and another two unpublished robust RT-qPCR assays, due to their superior sensitivity and repeatability (WOAHA, 2022). Lastly, various cell lines, including E-11 (Eyngor et al., 2014), OMB (Kembou Tsoufack et al., 2017), *Pristolepis fasciatus* caudal fin (CFF) (Behera et al., 2018), Nile tilapia brain (OnIB) (Thangaraj et al., 2018), and others, have been employed for TiLV isolation.

3. Prevention and control strategies for viral infections

Considering that tilapia is a commercially important protein source, losses in production due to viral outbreaks may have significant negative socio-economic impacts and compromise food security (Ansah et al., 2014; Wang et al., 2023). To mitigate these impacts, effective strategies for preventing and controlling viral disease epidemics need to be developed and implemented in aquaculture farms. One such strategy involves the use of antiviral therapeutics or prophylactics. However, presently, there is a lack of commercially available therapeutics to treat viral diseases in tilapia aquaculture (Jansen et al., 2019; Pereiro et al., 2021). Although past studies have developed therapeutics for viruses like LCDV (Harikrishnan, Kim, et al., 2010; Sun et al., 2013), IPNV (Ooi et al., 2008; Ren et al., 2022), NNV (Gopiesh Khanna et al., 2011; Kuo et al., 2016), ISKNV (Deng et al., 2022), and other megalocytiviruses (Chen et al., 2013), none have been certified for commercial use. Furthermore, these therapeutics have only been tested experimental conditions, leaving uncertainty regarding their performance in field applications. Notably, tilapia was not the model fish used in these studies, posing a challenge, as certain therapeutics (e.g., recombinant type I interferons [IFNs]) derived from other fish species may require specific receptors to be effective, potentially rendering them ineffective in tilapia (Pereiro et al., 2021). Due to these limitations, strategies such as good management practices (GMPs), biosecurity measures, and surveillance programs remain the primary methods for preventing and controlling viral outbreaks (Jansen et al., 2019; Kibenge et al., 2012).

Stress induced by inadequate culture conditions or during fish transportation often increases the vulnerability of fish to viral pathogens (Reverter et al., 2021; Wang et al., 2023), emphasizing the importance of GMPs to mitigate stress and reduce viral outbreak incidences (Mugimba et al., 2021; Oliva-Teles, 2012). Maintaining optimal water quality, appropriate fish stocking densities, effective sanitization practices, and providing adequate nutrition, collectively contribute to minimizing stress levels in cultivated fish (Ferguson et al., 2014; Mugimba et al., 2021; Wang et al., 2023). Biosecurity measures, in conjunction with screening and surveillance programs, play a crucial role in controlling disease spread. These measures encompass stringent quarantine protocols, prompt disposal of moribund fish, and the

establishment of different zones within aquaculture facilities (Assefa & Abunna, 2018; Johan & Zainathan, 2020; Mugimba et al., 2021). Biosecurity also entails rigorous sanitation and disinfection practices at all farming stages to prevent viral transmission (Mugimba et al., 2021). Several disinfectants have demonstrated efficacy in reducing viruses: TiLV (Jaemwimol et al., 2019; Soto et al., 2019), ISKNV (Fusianto et al., 2019), IPNV (Elliott & Amend, 1978), NNV (Volpe, Errani, Zamparo, & Ciulli, 2023), and LCDV (Cano et al., 2013) in *in vitro* or *in vivo* experiments, suggesting their potential application in aquaculture settings. However, further investigations are necessary to assess the efficacy of these disinfectants under field conditions and their potential impact on cultivated fish and the environment.

Given the propensity of viral diseases affecting tilapia for both horizontal and vertical transmission, it is imperative for tilapia farmers to procure specific pathogen-free (SPF) stocks to minimize the introduction of viral pathogens into their farms (Jansen et al., 2019; Machimbirike et al., 2019). Previous studies have demonstrated that different tilapia strains exhibit varying degrees of resistance to viral diseases such as TiLV (Adamek et al., 2022; Ferguson et al., 2014). Moreover, resistance to TiLV infection has been established as heritable (Barría et al., 2020), indicating the feasibility of selective breeding to enhance resistance to TiLV and other viral diseases, thereby mitigating the risk of epidemics (Houston, 2017).

To expedite the development of disease-resistant strains, marker-assisted selection (MAS) can be employed to identify quantitative trait loci (QTL) associated with disease resistance (Mugimba et al., 2021; Wang et al., 2023). Recently, a QTL linked to TiLV resistance in tilapia was pinpointed, offering a promising avenue for utilizing MAS in selecting TiLV-resistant tilapia for breeding programs (Barría et al., 2021). Meanwhile, early detection of viruses in aquaculture can facilitate the implementation of timely control measures to contain the spread of disease (Oidtmann et al., 2011; Stärk et al., 2019). Both biosecurity and surveillance in aquaculture are bolstered by the utilization of various diagnostic tools, including viral isolation, histopathology, immunoassays, and molecular-based methods (Mugimba et al., 2021). The accurate diagnosis of viral diseases in tilapia is particularly crucial, as many of these diseases exhibit overlapping clinical signs, making it challenging to differentiate between different viral infections based solely on gross lesions. Among the available diagnostic tools, molecular-based methods such as PCR are frequently employed for their speed, sensitivity, and specificity (Dong et al., 2023; Prasartset & Surachetpong, 2023).

However, the use of molecular-based diagnostics often requires well-equipped laboratories and trained personnel, resources that may not be readily available in every country (Minardi et al., 2019; Yin et al., 2019). For instance, several African countries lack the necessary laboratory infrastructure and skilled personnel, leading to delays in diagnosing TiLV and implementing control measures (Hounmanou et al., 2018). To address this challenge, the development of rapid diagnostic kits capable of on-site testing without the need for sophisticated equipment or expertise is essential for the timely detection of current viral infections affecting tilapia (Minardi et al., 2019; Wang et al., 2023).

Alternatively, vaccines represent another prophylactic strategy, wherein their administration in fish stimulates an immune response against specific pathogens (Ma et al., 2019). Several vaccines against TiLV (Mai et al., 2021; Yu et al., 2022) and ISKNV (Throngnumchai et al., 2021) have been developed for tilapia, with only one vaccine for ISKNV (AQUAVAC® Iridov) currently commercialized (Wang et al., 2023). Additionally, vaccines for IPNV (Munang'andu et al., 2014), NNV (Vimal et al., 2016), and LCDV (Zheng et al., 2010) have been developed for other fish species, with commercialized vaccines available for IPNV and NNV (Assefa & Abunna, 2018; Ma et al., 2019). However, further research is needed on the development of IPNV, NNV, and LCDV vaccines specifically tailored for tilapia, as existing vaccines may not perform optimally due to differences in adjuvants, dosage, delivery route, and required vaccination frequency (Wang et al., 2023).

Although vaccines have the potential to reduce losses caused by viral diseases (Ma et al., 2019), several factors must be considered. One such factor is the cost-effectiveness of vaccines, which should be ensured to promote their widespread use in tilapia farms worldwide (Jansen et al., 2019; Kembou-Ringert et al., 2023). It is especially important to ensure affordability for small-scale tilapia farmers (Hounmanou et al., 2018; Kembou-Ringert et al., 2023). Nonetheless, vaccination cannot be applied to tilapia larvae or fry due to their underdeveloped immune systems (Machimbirike et al., 2019). This limitation is significant as many viral diseases, such as TiLV, TLEV, NNV, and ISKNV, infect these early life stages of tilapia (Dong, Siriroob, et al., 2017; Keawcharoen et al., 2015; Shlapobersky et al., 2010; Subramaniam et al., 2016).

Instead, immunostimulants could potentially be used to enhance the innate immune responses of young tilapia against viral infections (Bricknell & Dalmo, 2005; Huttenhuis et al., 2006). Indeed, Elkatatny et al. (2020) demonstrated upregulation of various immune-related genes (e.g., Interleukin-1 β [il-1 β]) when Nile tilapia fry were administered an immunostimulant consisting of an amino acid mixture. Furthermore, the administration of immunostimulants, including yeast (Abdel-Tawwab et al., 2008), extract of custard apple leaf (Almarri et al., 2023), garlic, and echinacea (Aly & Mohamed, 2010), has been shown to improve the immune response of adult or juvenile tilapia under experimental conditions. Similarly, enhanced immune responses were observed after incorporating probiotics such as *Bacillus pumilus* (Aly et al., 2008) and *Lactobacillus rhamnosus* (Pirarat et al., 2011) into the tilapia diet. Specifically, diets supplemented with *Bacillus* spp. resulted in lower mortality rates and upregulation of immune-associated genes during experimental TiLV infection in red hybrid tilapia (*Oreochromis* spp.) (Waiyamtira et al., 2020). Additional studies also reported reduced mortality rates caused by NNV (Xiao Joe et al., 2021), LCDV (Harikrishnan, Balasundaram, & Heo, 2010), and iridovirus (Liu et al., 2012) under experimental conditions in other fish species fed with probiotics. Thus, immunostimulants and probiotics show promise in limiting viral diseases. Nevertheless, field application studies and cost-benefit analyses should be conducted to determine the feasibility of this strategy (Aly & Mohamed, 2010; Wang et al., 2023).

Overall, viral disease management in tilapia aquaculture requires the implementation of various strategies, as no single strategy is sufficient (Assefa & Abunna, 2018). Moreover, these strategies must be sustainable, practical, and cost-effective for tilapia farmers to effectively reduce viral infections (Jansen et al., 2019).

4. Conclusion

The emergence of viral infections in tilapia aquaculture is inevitable due to the widespread intensification of production. Among viral diseases, TiLV has received extensive research attention due to its current global impact on tilapia aquaculture. In contrast, limited follow-up research has been conducted on other viral infections affecting tilapia. Consequently, the genetic information, pathogenesis, epidemiology, and distribution of these viral diseases are yet to be fully determined, leaving the true impact of these diseases on tilapia aquaculture uncertain.

All life stages of tilapia are susceptible to varying viral diseases, posing significant risks to all stages of tilapia production. However, the incidence of viral outbreaks can be reduced by acquiring sufficient knowledge of viral diseases to implement effective prevention and control measures. Biosecurity and surveillance programs are the most effective methods to curb viral diseases, but efforts should be placed on the development of vaccines and therapeutics to further mitigate the losses. Furthermore, future studies should also prioritise investigating the lesser-known viral diseases, as the lack of awareness and study of these diseases in tilapia may inadvertently lead to epidemics in tilapia production.

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CRediT authorship contribution statement

Christal Winona Clyde: Writing – review & editing, Writing – original draft, Methodology, Data curation, Conceptualization. **Jin Ping Tan:** Writing – original draft, Methodology, Investigation. **Swee Keong Yeap:** Writing – review & editing, Validation, Supervision, Funding acquisition, Conceptualization. **Chean Yeah Yong:** Writing – review & editing, Validation, Supervision, Funding acquisition, Conceptualization.

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During the preparation of this work the author(s) used Chat GPT 3.5 and Perplexity Labs in order to correct grammar. After using this tool, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the publication.

Declaration of competing interest

The authors declare no conflict of interest.

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